

also 'returned without review' (sent back to applicants without a score), according to data shared with *Nature* by Grant Witness, a watchdog project that tracks changes to research funding. This sparked concerns that the type of science that the NSF would fund had also changed.

But some of these worries have been allayed with the announcement of a record-breaking number of GRFP awardees this year.

Brian Stone, who is standing in as NSF director until a permanent one is confirmed, said in a statement that the continuation of the programme reflects the Trump administration's "strong focus on building talent and investing in individual researchers". He added: "I'm excited to see how these emerging STEM [science, technology, engineering and mathematics] leaders will shape the future."

The agency did not respond to *Nature*'s queries about changes to the programme or applications returned without review. On its website, the NSF states that "the number of applications returned without review this year has not changed substantially since last year".

Award winners

Almost 14,000 young researchers applied for a 2026 GRFP award, submitting a research plan and personal statement that were reviewed by an independent panel of researchers. Typically, only about one in every six applicants receive one of the prestigious fellowships.

In addition to covering tuition fees, the fellowships come with an annual stipend of US\$37,000 for three years. Since 1952, when the GRFP began, it has supported more than 70,000 researchers, and at least 40 of those have gone on to receive Nobel prizes.

After rumours swirled that the Trump administration would request a massive cut to the NSF's approximately \$9-billion budget for the 2026 fiscal year, in April last year the agency cut in half the approximately 2,000 GRFP awards usually handed out. Months later, however, 500 awards were added – mostly in areas in which the Trump administration wants the United States to be a world leader, including artificial intelligence and quantum science – bringing the total to 1,500.

In February this year, NSF leaders announced at a board meeting that they intended to reshape the agency to fund more research on quantum science and AI, something reflected in the newly announced awards. Fifty-three of the latest GRFP awards are categorized as being for quantum science, a 39% increase from the previous year, and 103 are listed under AI or machine learning, a 17% increase.

Fields funded by the agency's engineering directorate saw the largest boost in GRFP awardees in 2026, from 406 (or 27% of the total) last year to 914 (or 35% of the total) this year.

But there were gains in other fields too. Research funded by the biological sciences

directorate rebounded, going from 214 awardees (or 14% of the total) last year to 486 (or 19% of the total) this year. Many researchers had been particularly worried about the biological sciences, given the reorientation of the NSF to focus on quantum and AI. With the exception of 2025, biological sciences applicants have received between 21% and 27% of the GRFP awards during the past decade.

After the awards were announced, many of the recipients from the record-breaking cohort took to the Internet to celebrate.

"Beyond grateful and still in disbelief," posted Lena Kemmelmeier, a psychology PhD student at the University of California, San Diego, on the social-media platform Bluesky. "Thank you to my wonderful lab mates."

Another student, posting anonymously on the social-media platform Reddit, where many GRFP hopefuls have commiserated, shared their excitement: "im a first gen PhD student and this is such a huge accomplishment for me!! Im tearing up in the middle of writing this and my hands are shaking!"

HUMAN SCIENTISTS BEAT THE BEST AI AGENTS ON COMPLEX TASKS

State-of-the-industry report finds that, despite limitations, researchers have embraced AI systems.

By Nicola Jones

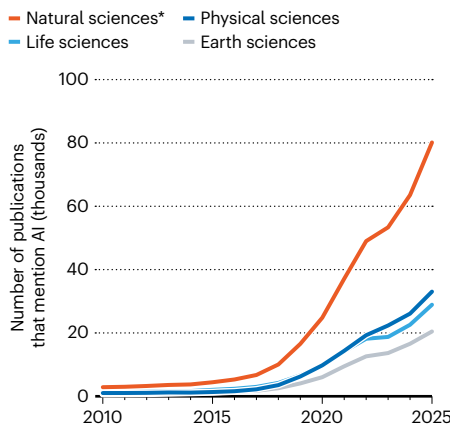
In an indication of how quickly scientists are embracing artificial intelligence, the number of publications in the natural sciences that mention AI grew almost 30-fold from 2010 to 2025, according to an influential annual report.

The proportion of publications in any given natural-sciences field that mention AI ranges from 6% to 9% (see 'AI paper boom'), according to the Artificial Intelligence Index Report 2026, released last week by the Stanford Institute for Human-Centered AI at Stanford University in California (see go.nature.com/4ctbkrq).

"Scientists have really embraced this AI era,"

AI PAPER BOOM

The number of natural-science publications – papers, preprints and other material – that mention artificial intelligence has increased sharply over the past 15 years. The natural sciences are comprised of physical, life and Earth sciences.



*Totals do not tally as some publications are included in more than one sub-field.

says computer scientist Yolanda Gil at the University of Southern California in Los Angeles, who led this year's index report.

Alongside the boom in AI-related science publications, the report also lists a host of newly released science foundation models – AI models that are broadly trained to take on a wide range of tasks, but also specially trained on massive data sets from a specific domain of science.

Many researchers have started to rely on AI 'agents' that autonomously carry out actions including scientific workflows, but the report is sceptical about the agents' performance. AI agents still struggle to perform multistep workflows reliably, the report says, with the best AI agents scoring roughly half as well as human specialists with PhDs. "Agents are wonderful, but we are still far from a place where we understand how to use them effectively," says Gil.

Growing numbers

The report says that in 2025, more than 80,000 papers, preprints and other types of publication in the natural sciences – which include life, physical and Earth sciences – mentioned AI, 26% more than in 2024. Physical sciences had the most publications that mention AI, with 33,000. The Earth sciences category had the highest percentage, at 9%.

The boom in AI-related science papers is "not surprising", says computer scientist Arvind Narayanan at Princeton University in New Jersey. But it's not clear, he says, whether the rise of AI use is productive for science. "Whether or not this explosive growth is meaningful is hotly debated," Narayanan says. "My

News in focus

view is that it is happening too fast, without giving scientific norms time to adjust, and so the quality of research has taken a nosedive.”

Gil says there isn't much evidence yet that AI is improving scientists' productivity. “The studies are limited,” she says. But, she adds, scientists “can't live without it. If you took AI away from them, there would be a riot. So it must be helping in some way.”

Top models

As uptake has grown, so has the number of AI platforms available to researchers. The past year has seen the emergence of many science foundation models, including the first one for astronomy: AION-1, which was trained on more than 200 million celestial objects. This training helps it to classify galaxies and to estimate their properties. “When I talked to scientists in 2024 and said ‘There's foundation models for science’, scientists would not know what that means. They didn't know they existed. I think we have seen that really advance very quickly,” says Gil.

The performance of general-purpose foundation models, which include large language models (LLMs) such as those used by chatbots, on expert-level tests continued to improve rapidly in 2025, the index says. “I keep thinking: next year it's going to plateau. And it doesn't,” says Gil. But these leading AI models still make basic mistakes, such as reading analog clocks incorrectly 50% of the time.

On a positive note, models that generate videos “are beginning to learn how the physical world actually works”, the index says. The AI community has long hoped that models will be able to learn the basic rules of physics, such as how gravity affects a bouncing ball. In 2025, Google DeepMind's Veo 3 video model was able to simulate phenomena such as buoyancy without being trained to do so. But there's a long way to go, says Narayanan. “Despite some minor successes,” he says, “the dream seems pretty far away.”

Agentic abilities

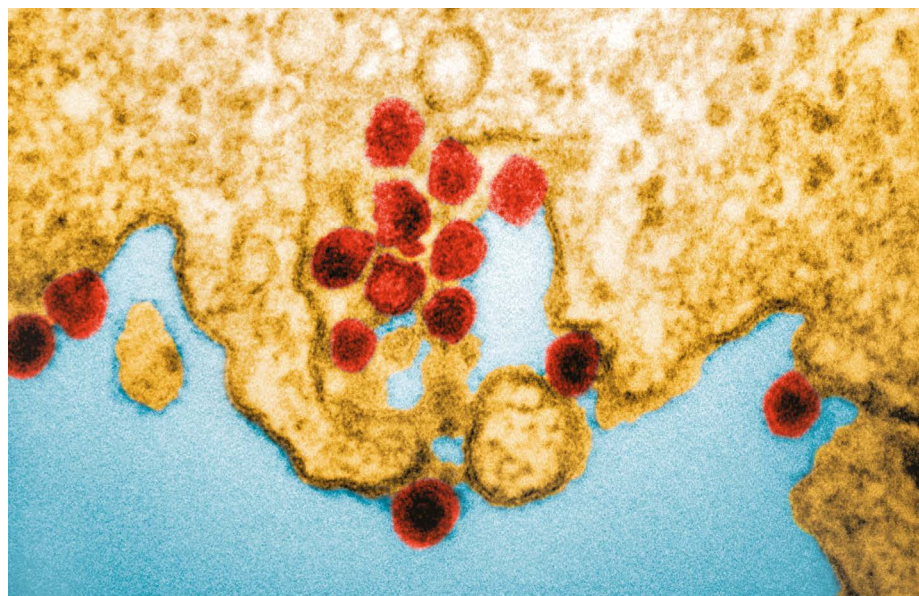
Several benchmark tests emerged in 2025 to score the performance of AI agents on science-relevant tasks. A test called Paper-Arena, for example, gives LLM-based agents a research question and then evaluates their ability to “formulate a reasoning plan, interact with multiple papers and invoke appropriate tools to produce a well-grounded answer”, according to an unreviewed preprint describing that benchmark (D. Wang *et al.* Preprint at arXiv <https://doi.org/q3fw>; 2025). On this test, even the best LLM powering a skilled agent achieved an accuracy of just 39%.

Narayanan has a more positive view of what agents. In November, his own team showed that agents can ace one benchmark test for reproducing computational results in scientific papers, he says. “There is obviously

a lot of hype around agents, but they are already extremely useful to researchers as time-saving tools.”

The use of AI agents is part of a general trend of scientists using AI tools not just for

data analysis but also for broader attempts at hypothesis generation and discovery. That's exciting, but the index concludes that, for now, “the list of experimentally confirmed AI discoveries remains short”.



Chikungunya particles (red) infecting a cell.

O. SCHWARTZ, M. SOURISSEAU, MC. PREVOST, INST. PASTEUR/SPL

VIRUSES ALLEGEDLY STOLEN FROM HIGH-SECURITY LAB

The samples have been recovered by Brazilian police, but questions linger over motive.

By Mariana Lenharo

Brazil's federal police arrested a researcher late last month for allegedly taking samples of viruses from a high-security biosafety laboratory at a leading Brazilian university. Soledad Palameta Miller, a virologist at the State University of Campinas (Unicamp), was released on bail on 24 March and faces charges including theft.

Police have recovered the missing samples – which included specimens of the viruses chikungunya, dengue and Epstein–Barr, according to the Brazilian news programme Fantástico – from other locations on Unicamp's campus. Brazil's National Health Regulatory Agency (Anvisa) said this month that it had evaluated the recovered samples and that they did not pose a risk to people's health.

But the news has caused a stir in the Brazilian virology community, whose members are

wondering how the lapse could have happened at a lab designated biosafety-level-3 (BSL-3), the second-highest security classification.

The community is “perplexed”, says Paulo Sanches, a virologist at São Paulo State University in Araraquara, Brazil. “No sample can be removed from a lab with this biosafety level without authorization.” The news comes as Brazil is building support for the construction of its first BSL-4 lab – the highest level of biosafety facility – only a few kilometres from Unicamp.

Miller did not respond to a request for comment from *Nature*. In a statement, Unicamp said that it is cooperating fully with the police and is conducting an internal investigation.

Police investigation

The samples were allegedly stolen from Unicamp's Laboratory of Virology and Applied Biotechnology. BSL-3 facilities are equipped with air filters and containment systems to